From Perception to Punchline:
Empowering VLM with the Art of In-the-wild Memes

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HUMOR

Hierarchical Understanding and Meme Optimization

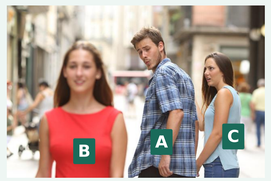
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1a What makes a caption a meme?

Meme generation goes beyond captioning: a shared visual relation supports many *context-specific punchlines*.

Template + roles



Technology / Apple fanatic



A · Subject	B · New focus	C · Neglected
APPLE FANS	IPHONE 11	IPHONE 10

Reusable template intent: infatuation with the new and abandonment of the old.

Input: template + user intent. Output: humorous text assigned to designated image regions.

A testbed for subjective multimodal generation.

1b Learn preferences, not global scores

Humor is **context-dependent**. Global scores assume a common scale across unrelated jokes; within-template pairs share a **coherent semantic structure**.

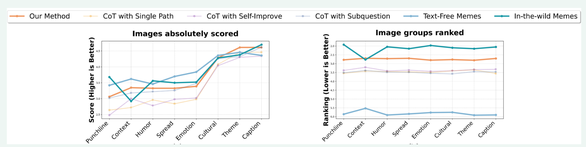


Fig. 4: absolute scores blur real vs. text-free memes; local ranks separate them more clearly.

Human check: ρ = 0.72; τ = 0.63

9 annotators · 5 groups · 30 memes (App. K.5)

5 pair types: wrong text / location / boring / edits / generated.

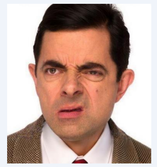


Original | Wrong text | Wrong position

2a Explore multiple scenes, then reverse-anchor

Single-answer rationales expose only one route. HUMOR separates *template-level intent* from *context-level grounding* (§3.1).

1 MULTI-SCENARIO ASSOCIATION



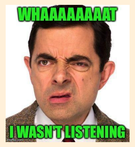
Bewilderment / annoyance

→ A confusing work problem → bewildered reaction

→ A bad joke → exaggerated “painful” response

→ Social ignorance → refusing to listen

Trace back to the compatible scene



Real reference

2 REVERSE-ANCHOR FROM A REAL MEME

Gold caption c → infer U: socializing / ignorance, then trace back to a compatible path; retain other associations.

$$\tilde{r} = \arg\max_{r} P_\phi(c \mid r, I, U)$$

INFERENCE Explore and select internally; no gold caption is available.

Fig. 14 / App. B; scenario summaries paraphrase the paper’s traces.

2b Template-local reward → group-wise optimization

Compare memes **only within G** (a shared template). Train a reward model on **constructed preference pairs**, rather than calibrating global humor (§3.2).

PAIRWISE MODEL · BCE TRAINING

$$p_{ij}^G = \sigma[s_\phi(m_i) - s_\phi(m_j)]$$

$$\ell = -y \log p - (1-y)\log(1-p)$$

EXPECTED BORDA COUNT

$$\mathrm{EBC}_G(i) = \sum_{j \neq i} p_{ij}^G$$

GROUP-LOCAL AGGREGATION · Appendix E, Algorithm 1

1 Estimate pairs inside the template group; omit missing edges.

2 Sum expected wins; normalize by observed opponents if sparse.

3 Rank candidates; optional qG = softmax(EBC / T), T > 0.

POLICY UPDATE · reference = CoT policy (§3.3)

$$\mathcal{L} = -\mathbb{E}[\sum_k q_G(m_k) \log \pi_\theta(c_k \mid I)] + \beta\, \mathbb{E}[\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{CoT}})]$$

$$H(\pi_\theta) \geq H(\pi_{\mathrm{CoT}}) - \sqrt{\Delta_{\mathrm{KL}}/2}$$

KL bounds potential degradation; it does not guarantee gains.

3 Evidence: generation, reward & transfer

30 native-English raters · 3–5 memes / method / rater. Human scores: 0–5 ↑; tuned variants use the same 7B.

Humor: 2.39 → 2.68 with HUMOR-CoT

Single-path: 1.87 | RL preview: 2.83

Model / variant	Hum.	Read.	Rel.	Orig.	Dist.
Qwen2.5-7B	2.39	3.35	2.91	2.57	0.564
Qwen2.5-32B	2.54	3.52	3.09	2.76	0.566
InternVL3-8B	2.39	2.79	3.04	2.79	0.564
GLM-4.1V-9B	1.73	2.62	2.75	2.71	0.572
Keye-VL-8B	2.35	3.19	2.99	2.71	0.580
GPT-4o	2.70	2.99	3.21	**2.97**	0.552
Gemini-2.5-flash	2.81	3.29	3.25	2.88	0.561
HUMOR-CoT	2.68	**3.70**	3.50	2.90	0.590
CoT: Single Path	1.87	2.79	2.68	2.45	0.570
CoT: Self-Improve	2.38	3.68	3.00	2.65	0.578
CoT: Subquestion	1.85	3.32	2.58	2.47	**0.597**
HUMOR-RL · preview	**2.83**	3.67	**3.55**	2.79	0.588

Table 1 · Bold: column best. Dist.: context-swap distance ↑.

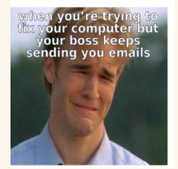
RL trade-off: humor rises to 2.83, while originality drops **2.90 → 2.79**. RL remains a preview.

Single-path CoT



Literal work scenario

HUMOR-CoT



Specific conflict: the boss interrupts the repair by email.

Fig. 16 · same template / work intent; original outputs.

Reward–human agreement

100 MaxDiff raters · 5 templates · Kendall τ ↑

	G1	G2	G3	G4	G5
Qwen 7B · base	0.16	0.28	0.47	-0.10	0.29
Qwen 7B · trained	0.47	0.56	0.42	0.14	0.47
Qwen 32B · base	0.16	0.16	-0.02	0.14	0.29
Qwen 32B · trained	0.29	0.47	0.07	0.30	0.42
Keye 8B · base	0.05	0.09	0.16	0.29	0.16
Keye 8B · trained	**0.78**	**0.77**	**0.78**	**0.78**	**0.78**

20 unseen templates · Gemini mean rank ↓

	Img–text	Theme	Emo-tion	Humor	Punch-line	Cultural fit	Con-text	Spread
Human	1.2	2.2	1.2	1.4	1.0	1.0	1.0	1.0
HUMOR-CoT	**2.8**	**2.4**	**2.6**	**3.6**	**3.4**	**2.4**	**2.4**	**2.8**
Qwen 32B	4.2	4.2	4.2	4.0	4.2	4.2	4.2	4.0
Keye	4.2	4.4	4.0	4.4	4.2	3.8	4.0	4.4
InternVL	4.4	4.0	4.4	4.6	4.8	5.0	5.0	4.8
Qwen 7B	5.2	4.6	5.2	4.2	4.8	5.6	5.6	5.0
GLM	6.0	6.2	6.4	5.8	5.6	6.0	5.8	6.0

Scope: English meme culture · limited human samples · ~2–3× output tokens · concrete keywords can induce literalism.

3,345 train / 368 test | karroyan13@gmail.com